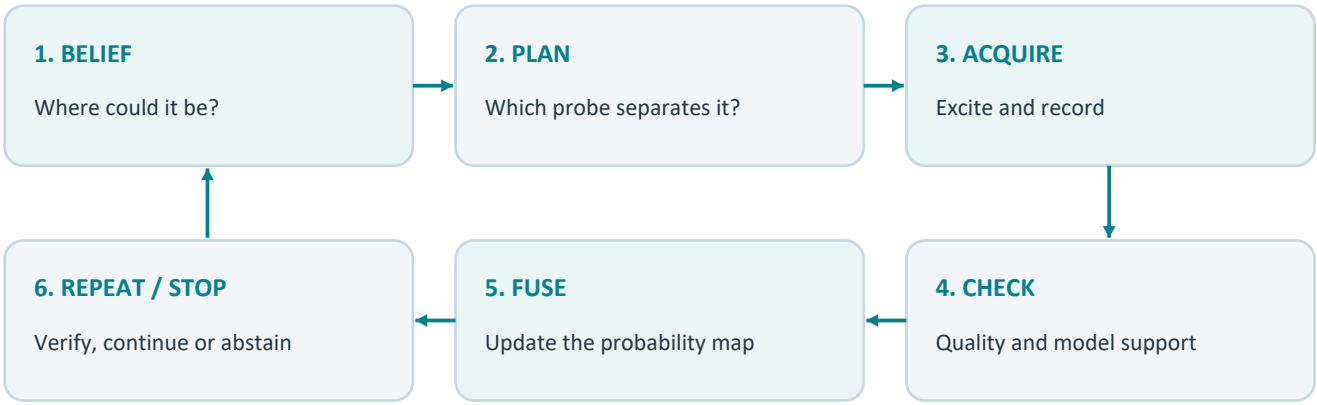
01 / REVIEW 1 STUDY GUIDE

ARGUS

Understand the project. Explain the physics. Defend the evidence.

Adaptive Recursive Guided Uncertainty Sensing
A local research prototype that chooses what acoustic experiment to perform next while locating one dominant hidden defect.



Your anchor sentence
ARGUS maintains a probability map of a hidden defect, predicts how rival locations would respond to possible probes, selects an informative feasible experiment, and updates the same belief with the resulting waveform.

Prepared: 7 October 2026 | **Review:** 8 October 2026
Basis: current local code, saved artifacts, and fresh software checks.
General technical Review 1 preparation; course and review duration were not supplied.

Source trail: Working tree inspected as provided, including existing uncommitted changes. No implementation files were changed for this guide.

02 / STUDY ROUTE

How to use this guide

You do not need to memorize every file tonight. Learn the measurement loop, the ellipse explanation, the Bayesian update, and the evidence boundaries first. Then practise explaining them without reading.

Read first	What you should be able to say
Pages 3-6	Problem, architecture, stack, and one request through the code
Pages 7-10, 14-18	Wave physics, time of flight, Bayesian inference, and experiment selection
Pages 19-24	Calibration, OOD, experimental policies, ML, and public data
Pages 27-32	Results, live demo, review script, and viva practice
Pages 33-35	Limits, next work, glossary, and source trail

Reading map

Topic	Pages	Topic	Pages
Problem and software	3-6	NEO, calibration and trust	19-22
Core wave geometry	7-10	ML and measured data	23-24
Virtual Lab and signals	11-14	Hardware and interface	25-26
Belief and planning	15-18	Evidence and presentation	27-32

A 90-minute preparation sequence

- **First 25 minutes:** say the 30-second explanation; draw source, defect, receiver, and the closed loop.
- **Next 25 minutes:** solve the worked time-of-flight and Bayes examples. Explain entropy, MAP, calibration, and counterfactual planning.
- **Next 20 minutes:** rehearse the actual demo and the result statements. Keep experimental options off for the default walkthrough.
- **Final 20 minutes:** answer viva questions aloud. Finish with a realistic physical-validation milestone.

Three mistakes to avoid
Do not present simulated error as field accuracy; do not call a response-prediction R-squared an accuracy percentage; do not describe the 3D probability relief as actual defect depth.

03 / PURPOSE

The problem and the proposed solution

Imagine an opaque panel with a cavity, poorly bonded region, or delamination hidden inside. You can excite the panel and record vibration, but you cannot directly see the interior. The question is both **where the defect might be** and **where to probe next**.

A fixed inspection schedule collects predetermined measurements. A passive classifier interprets measurements chosen by someone else. ARGUS makes acquisition itself adaptive: its present uncertainty influences the next source position, receiver position, frequency band, and waveform.

Review-ready problem statement

> **Problem statement**
> Given a limited measurement budget and uncertain wave propagation, choose sequential feasible acoustic experiments that help localize one dominant hidden scatterer while recording uncertainty and acquisition cost.

Project objectives

- Maintain a normalized spatial belief that combines measurements over time.
- Use an interpretable physical response model to predict candidate outcomes.
- Choose experiments that separate competing hypotheses after movement, energy, time, and repetition penalties.
- Reject or downweight poor evidence, request calibration when appropriate, and verify a concentrated result before a confident stop.
- Expose waveforms, score components, history, and evidence records so decisions can be examined.

What is produced

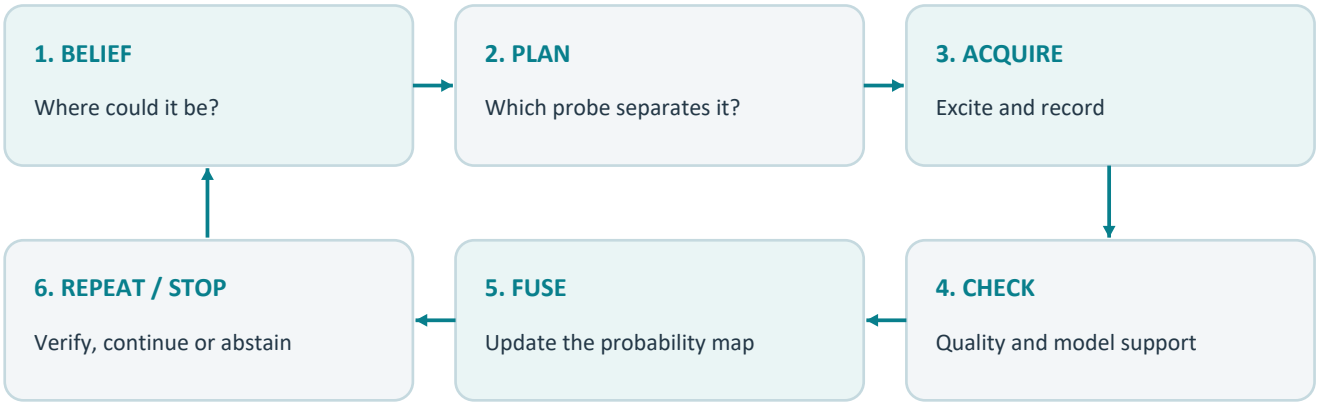
The output is a probable location and uncertainty summary, plus a recommended next experiment and its rationale. In simulation, the hidden truth can be revealed for scoring. On a physical object, the system has no automatic access to ground truth.

Potential application areas include panels and bonded structures. These are application motivations; the present repository does not establish deployment readiness in any specific industrial structure.

Source trail: [R1] README; [R2] current technical disclosure; [R3] engine and domain models.

04 / SYSTEM ARCHITECTURE

The complete measurement loop



Step	What the system actually does
Initialize	Create a session, panel geometry, propagation prior, uniform grid, and initial recommendation.
Plan	Select an action mode, generate source/receiver/waveform candidates, enforce constraints, predict rival outcomes, and rank utilities.
Acquire	Generate a seeded synthetic response or accept a WAV/microphone/serial/phone signal with experiment metadata.
Check	Validate samples and quality. For diagnostic evidence, assess model discrepancy and OOD against the pre-measurement belief.
Infer	Reference actions update metrology. Accepted diagnostic evidence updates the location grid through a quality-tempered likelihood.
Persist	Store the waveform and state transition, append an evidence record, refresh the recommendation, and expose the result to the UI.
Decide	Continue, acquire calibration/verification, stop at budget/low value, or abstain under unsupported evidence.

Why it is a closed loop
The next experiment depends on the previous measurements. Planning and inference repeatedly change one another through the same session state.

Source trail: [R3] ArgusEngine; [R4] SessionManager; [R5] NeoExperimentPlanner; [R6] StoppingEngine.

05 / SOFTWARE

Technology stack and repository map

Technology	Role in this project
Python	Simulation, scientific computation, inference, planning, data preparation, and backend orchestration.
NumPy + SciPy	Array probability grids, geometry, filtering, correlation, FFT/PSD, resampling, and numerical integration.
FastAPI + Uvicorn	HTTP API and WebSocket endpoints; local service execution.
Pydantic	Validate session, experiment, device, and research request schemas.
SQLite	Persistent sessions, experiments, research jobs, events, and ledger records.
React 19 + TypeScript	Browser instrument, session UI, signal plots, camera/probe views, and type contracts.
Vinext + Vite	Current development/build runtime. package.json also lists Next.js; the actual commands use vinext.
PyTorch	Optional response-feature MLP training; CPU execution is available.
scikit-learn	Measured-data estimators, grouped evaluation, and simulation-to-reality experiments.
pyserial + sounddevice	ESP32 serial communication and local operating-system microphone access.
ESP32 / Arduino C++	Reference actuator/sensor firmware; GPIO drive control and buffered ADC acquisition.

Where to look

backend/app/: simulation, signal, inference, active_learning, control, digital_twin, ood, assurance, hardware, database, evidence, evaluation, replay, research, and services.
frontend/: app routes, components, hooks, services, utilities, and tests.
scripts/: launch checks, demos, generation, training, benchmarks, and physical-campaign tooling.
models/, datasets/, research_results/, benchmark_results/: data adapters, model metadata, and evidence artifacts.

Source trail: [R7] backend/requirements.txt; [R8] frontend/package.json; [R1] repository layout.

06 / CODE WALKTHROUGH

Follow one experiment through the code

When you press **Run Experiment**, React sends a request to the local API. The frontend is a view of authoritative backend state; it does not independently invent the posterior.

```
Browser button
  -> useArgusSession.run()
  -> POST /sessions/{id}/experiments/run
  -> SessionManager.run()
  -> ArgusEngine.run_recommended()
  -> process_signal()
  -> persist result + ledger + new recommendation
  -> JSON response -> updated React views
```

File / entry point	Responsibility
frontend/app/page.tsx	Launch controls and Overview, Reasoning, Signals, Evolution, Benchmark, Research, Evidence, and Registration views.
frontend/hooks/useArgusSession.ts	API calls, busy/error state, history refresh, and optional last-session identifier.
frontend/services/api.ts	API URL discovery and typed requests; checks service=argus before showing engine online.
backend/app/main.py	HTTP routes and WebSocket connections.
backend/app/services/session_manager.py	Session mode rules, validation, resampling, locking, persistence, and public state.
backend/app/services/engine.py	Acquisition-to-inference orchestration and action selection.

Important API examples

GET /health: readiness and service identity.
POST /sessions: create a simulation or physical session.
GET /sessions/{id}: current state and recommendation.
POST /sessions/{id}/experiments/upload: WAV acquisition.
GET /sessions/{id}/history: recorded experiment sequence.
POST /sessions/{id}/reveal: simulation truth only.

Source trail: [R4], [R8], [R9] API and session UI files.

07 / PHYSICS FOUNDATIONS

Why vibration reveals hidden structure

An exciter applies a changing force. The structure deforms, and elastic disturbances travel away from the source. A cavity or delamination changes local stiffness, mass, damping, or continuity, changing the recorded vibration through scattering, phase shifts, amplitude changes, or ringing.

Useful physical quantities

Quantity	Meaning and relevance
Frequency f (Hz)	Oscillations per second. Different frequencies can interact differently with a structural anomaly.
Wave speed v (m/s)	Converts propagation distance into arrival time. A wrong speed biases inferred position.
Wavelength lambda (m)	For a nondispersive wave, lambda=v/f. For a dispersive plate, use phase velocity at that frequency.
Attenuation alpha (1/m)	Amplitude loss with distance. This competes with defect strength and probe coupling.
Resonance and damping	Preferred oscillation rates and decay. Added mass or reduced stiffness can shift response; damping makes it fade.

Real plate waves and the project approximation

Real guided plate waves can have different symmetric and antisymmetric modes; their speed may vary with frequency. Composite directionality, boundaries, coupling, and thickness complicate propagation. COMSOL documents this modal and dispersive behavior for plate inspection [W2].

The main ARGUS backend uses one effective velocity, delayed copies of excitation, and an empirical scatter/ring model. It does not solve the elastic wave equation or reproduce full Lamb-wave dispersion. The separate Virtual Lab uses an ideal thin-plate bending approximation, explained on page 11.

Say this if challenged
Our model is physically inspired and interpretable. It captures useful geometry and timing, while physical material-specific wave calibration remains an experimental requirement.

Source trail: [R10] backend simulation; [R11] Virtual Lab; [W2] COMSOL guided-wave model.

08 / BISTATIC LOCALIZATION

One echo describes an ellipse

```
L(z) = distance(source, z) + distance(z, receiver)
t(z) = t0 + L(z)/v
L(z) = v * (measured arrival - t0)
```

The source and receiver are the ellipse foci. A single measured scattered arrival supports all points with approximately the same sum of distances. Noise turns the ideal curve into a band. This is why one waveform generally cannot identify a unique 2D point.

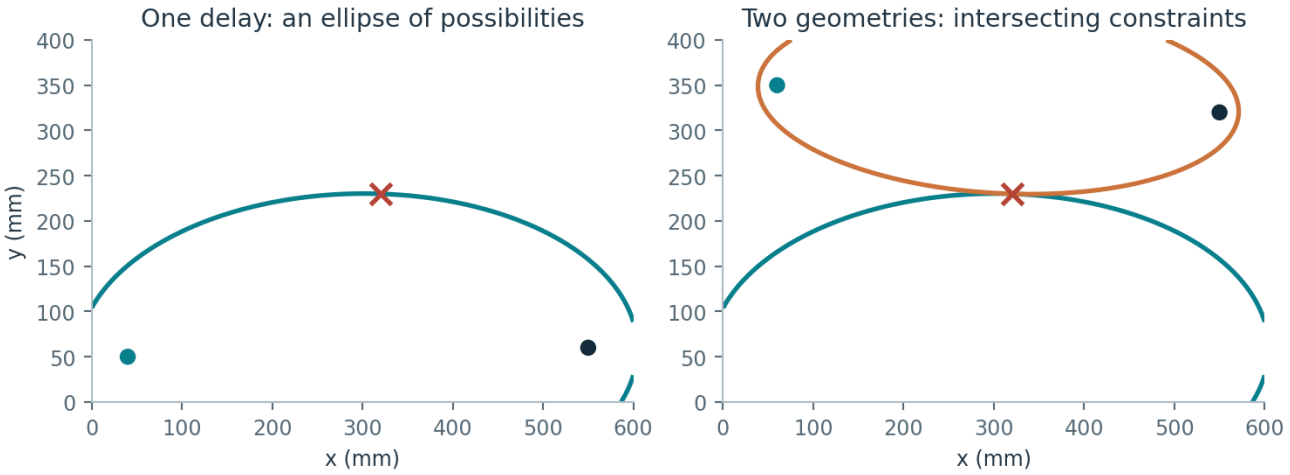


Figure 1. Analytic geometry illustration, not measured evidence. The red cross is the illustrative target. Different source/receiver arrangements change equal-path constraints.

A new geometry produces a different ellipse. Fusion increases belief where multiple observations agree, although ambiguity can remain. Several diverse geometries are generally more useful than repeating one geometric constraint.

If source and receiver coincide, the bistatic sum becomes twice the source-to-defect distance and the equal-delay curve becomes a circle. With separated probes, dividing total path length by two does not give the defect distance from either probe.

The physics-to-AI connection
Geometry converts a measured time delay into spatial evidence. The planner chooses geometry to make the remaining spatial explanations distinguishable.

Source trail: [R10] path_properties; [R12] measurement_likelihood; [R1] project knowledge.

09 / NUMERICAL EXAMPLE

A worked time-of-flight calculation

Use the main backend defaults for this example: a 0.60 m by 0.40 m panel, effective velocity 180 m/s, and instrument delay t0=0.0008 s. These are illustrative simulator parameters, not certified material constants.

Point	Normalized coordinates	Physical coordinates
Source S	(0.10, 0.20)	(0.060, 0.080) m
Receiver R	(0.90, 0.20)	(0.540, 0.080) m
Defect z	(0.50, 0.60)	(0.300, 0.240) m

```
x_m = x_normalized * 0.60
y_m = y_normalized * 0.40

|S-z| = sqrt(0.240^2 + 0.160^2) = 0.28844 m
|z-R| = 0.28844 m
L_scatter = 0.57689 m
t_scatter = 0.0008 + 0.57689/180 = 4.005 ms

L_direct = 0.48000 m
t_direct = 0.0008 + 0.48000/180 = 3.467 ms
Extra scattered delay = 0.538 ms
```

Sampling interpretation

At 16,000 samples/s, one sample spans 62.5 microseconds. The extra delay is about 8.61 sample intervals. ARGUS uses interpolation when simulating fractional delays; the spatial likelihood rounds predicted delays to sample lags and broadens correlation support.

Two common errors

- Computing distance in the normalized unit square ignores the rectangular panel scale. A normalized x step and the same y step represent different physical distances.
- The 62.5 microsecond sample interval is not a guarantee of localization precision. Waveform bandwidth, noise, geometry, coupling, calibration, and grid discretization also matter.

Source trail: [R3] physical_distance and defaults; [R10] fractional delay; [R12] lag lookup. Calculations are derived from these equations.

10 / REDUCED WAVE MODEL

The main backend digital twin

```
y(t) = A_direct * u(t-t_direct)
     + A_scatter * u(t-t_scatter)
     + damped_ringing(t) + noise(t) + drift(t)

t_direct  = t0 + |S-R|/v
t_scatter = t0 + (|S-z| + |z-R|)/v
A_direct  = exp(-alpha*|S-R|)/(1+4*|S-R|)
```

Here u(t) is the chosen excitation. The scatter gain combines defect-type sign, severity, normalized size, distance attenuation, geometric spreading, and a frequency-dependent resonance factor. The constants are reduced-model design choices.

```
A_scatter = type_factor * severity * size_scale
          * exp(-alpha*L)/(1+5*L)
          * resonance_gain * 0.43

ring(t) ~ sin(2*pi*f_ring*t) * exp(-damping*t)
```

Defect label in backend	Simulator reflection factor
cavity	+1.00
loose_region	+0.72
delamination	+0.86
dense_inclusion	-0.64

The sign models a possible polarity change; it is not a universal physical signature for these defect types. The localization path uses the magnitude of correlation, so this implementation is not a validated defect-type classifier.

The hidden acquisition simulator and the inference simulator are separate objects. Calibration changes the inference model without silently rewriting the hidden world that produces later synthetic observations. This allows tests of model mismatch.

What the twin means
A computational model predicts responses under proposed experiments. In this project the live twin is reduced order; calling it a digital twin does not imply validated finite-element fidelity.

Source trail: [R10] AcousticSimulator; [R3] ArgusEngine initialization; [R2] current support boundary.

11 / THIN-PLATE MODEL

The browser Virtual Lab has different physics

The /simulator route is a separate TypeScript teaching simulator. It lets users change material, thickness, geometry, defect, impact strength, noise, sensor response, and boundary reflectivity. These controls do not replace the main Bayesian backend model.

```
D = E*h^3 / (12*(1-nu^2))
b = sqrt(D/(rho*h))
omega = b*k^2  [ideal thin-plate bending wave]
c_phase = omega/k = sqrt(b*omega)
c_group = d(omega)/dk = 2*sqrt(b*omega)
lambda = c_phase/f = c_group/(2*f)
```

D is bending rigidity, E is Young's modulus, h is thickness, nu is Poisson's ratio, and rho is density. From these formulas, group velocity scales as sqrt(frequency) and sqrt(thickness) for a fixed material. The wavelength uses phase velocity.

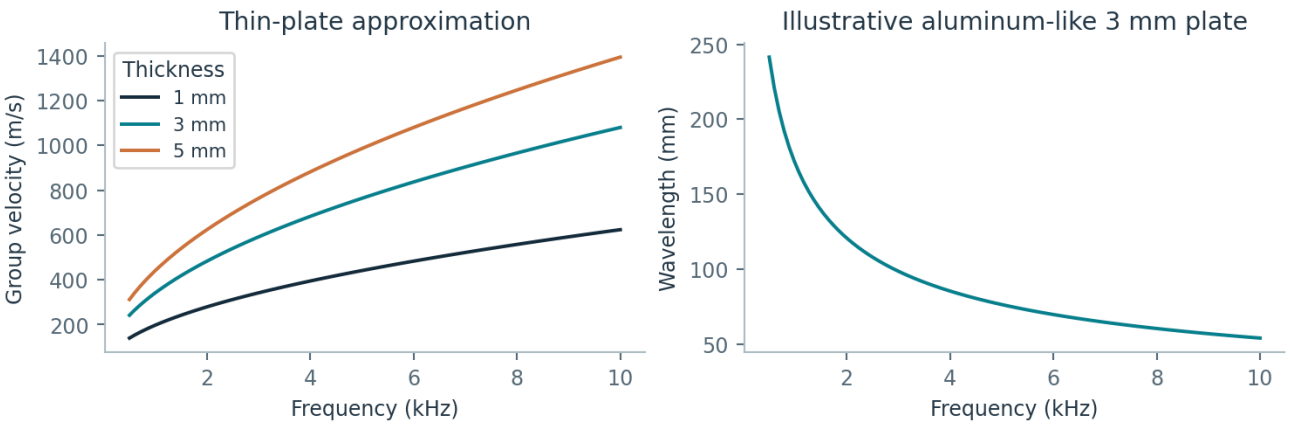


Figure 2. Calculated from the Virtual Lab thin-plate formulas and its illustrative aluminum parameters. This is theoretical scaling, not measured material characterization.

The Virtual Lab adds four image-source edge paths, size-to-wavelength scatter strength, approximate sensor transfer, damped carriers, and seeded noise. It approximates dispersion using a center-frequency speed and a time-varying carrier; it is not a full dispersive plate solver.

The ideal isotropic thin-plate assumptions should not be taken as a complete model of plywood or layered composites. Listed material values are demonstration presets rather than a calibrated specimen characterization.

Source trail: [R11] flexuralPlateProperties and simulateVirtualExperiment; [W3] COMSOL plate stiffness context. Dispersion derivation follows the implemented ideal model.

12 / EXCITATION DESIGN

Waveforms, bandwidth, and sampling

Waveform	Intuition and implementation
Chirp	Frequency sweeps over a band. Broad excitation gives a useful correlation signature and samples frequency-dependent response.
Sine / tone burst	A short narrowband oscillation. Useful for a selected frequency, but repeated cycles may give ambiguous timing peaks.
Impulse / Ricker	The backend impulse option is a smooth Ricker-like pulse, not an infinitely short mathematical delta.
Multisine	Several sinusoids combined; probes several frequencies in one emission.
Phase / complementary coded	Finite sign sequences modulate the carrier. Complementary-coded support does not itself establish ideal paired-code cancellation.
Spectrally notched	Selected frequency intervals are removed from a chirp-like waveform.

A Tukey window softens burst edges. The backend normalizes transformed waveforms before applying amplitude so coding or notching does not exceed the specified peak drive. The active burst is at most 10 ms; default recording duration is 120 ms.

```
Default fs = 16,000 Hz
Nyquist frequency = fs/2 = 8,000 Hz
Default recording = 0.12 s -> 1,920 samples
Configured maximum candidate frequency = 7,000 Hz
```

Nyquist is a sampling limit, not a hardware bandwidth specification. The planner enforces a frequency margin, amplitude and duration bounds, probe spacing, and no-go regions. Sampling and actuator capability must both be adequate in a physical setup.

Why choose the waveform too?
Moving probes changes geometric delays. Changing the waveform changes spectral interaction and correlation shape. ARGUS searches a finite bank of such combinations; it does not optimize every possible continuous waveform.

Source trail: [R10] excitation; [R13] configuration and constraints; [R5] waveform utility.

13 / SIGNAL PROCESSING

What the signal plots tell you

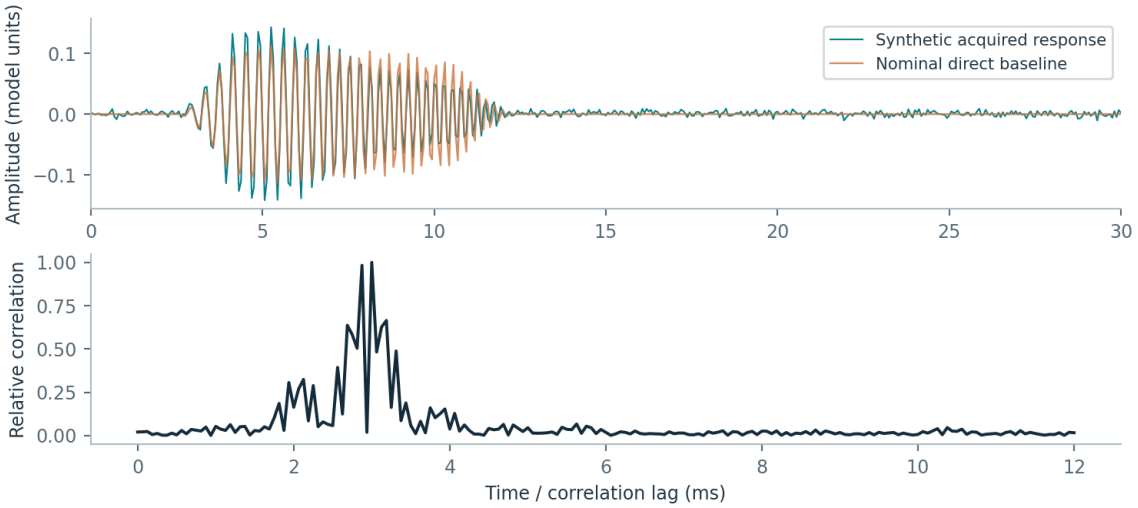


Figure 3. Current seed-17 synthetic response and a residual-correlation illustration. The displayed baseline uses the final calibrated inference material; this explanatory correlation plot is not a stored acquisition-time likelihood.

Operation / feature	Purpose
DC removal + bandpass	Remove the mean and emphasize the 250-7,000 Hz analysis band using a fourth-order Butterworth filter.
FFT + Welch PSD	Show frequency content and estimated power distribution.
Spectrogram	Show how frequency energy varies during the recording.
Hilbert envelope	Estimate an amplitude envelope, peak timing, and decay time.
RMS / peak / crest factor	Summarize amplitude and impulsiveness.
Spectral moments / band energy	Summarize spectral concentration and redistribution.
Noise / SNR estimates	Provide robust diagnostics from median-based estimates and signal magnitude.

The SNR and quality values are estimates, not laboratory measurements of sensor performance. Most displayed features support diagnostics and optional ML. The spatial update is specifically driven by baseline-subtracted matched filtering.

Code detail that matters
analyze_signal preprocesses the diagnostic plots and features. The present measurement_likelihood receives the original acquisition samples, subtracts its nominal baseline, and correlates the residual directly. Do not claim every inference sample passes through the displayed analysis filter.

Source trail: [R14] signal/processing.py; [R12] belief.py; [R3] process_signal. [W1] SciPy documents correlation-based matched filtering.

14 / LIKELIHOOD CONSTRUCTION

How a waveform becomes spatial evidence

A **likelihood** measures how compatible the new observation is with each candidate defect location. It is distinct from the posterior, which also includes the previous belief.

```
residual = measured_response - predicted_direct_baseline
C[k] = abs(sum_n residual[n] * excitation[n-k])
k(z) = round(fs * (t0 + L(z)/v))
score(z) = broadened correlation near k(z)
likelihood(z) ~ exp(effective_temperature * scaled_score(z))
```

Implementation sequence

- Compute a nominal direct response for the recorded experiment and remove it from the measured waveform.
- Use FFT-based full correlation with the known excitation; retain nonnegative lags and apply a seven-sample maximum filter.
- Compute bistatic delay for each grid-cell center and look up correlation support at that predicted lag.
- Smooth spatial scores with a Gaussian filter; use the 8th and 98th score percentiles for bounded scaling.
- Convert the bounded score to a positive exponential likelihood. Residual SNR tempers the effective sharpness.

Why matched filtering helps

A delayed copy of the excitation lines up at the correct correlation shift, producing strong support. Uncorrelated noise tends to align less consistently. In this code, absolute correlation also tolerates polarity reversal. SciPy provides this correlation operation [W1].

> **Scientific boundary**
> This is a generalized matched-filter likelihood, not a fully calibrated probability density of every possible waveform. A wrong baseline or arrival model can create a sharply concentrated but incorrect posterior.

Source trail: [R12] measurement_likelihood. [W1] SciPy correlate documentation. The spatial construction is repository-specific.

15 / RECURSIVE BELIEF

Bayesian inference without the mystery

```
posterior_next(z) = prior(z) * likelihood(z)^w
                  / sum_cells(prior * likelihood^w)

0 <= evidence weight w <= 1
```

The **prior** is the current belief before this measurement. The **likelihood** is new evidence. The **posterior** is the normalized belief afterward. At the next iteration, that posterior becomes the new prior. This recursion is the memory of the system.

Three-location teaching example

Location	Prior	Likelihood	Unnormalized	New posterior
A	0.50	0.10	0.050	0.125
B	0.30	0.90	0.270	0.675
C	0.20	0.40	0.080	0.200

The denominator is 0.050+0.270+0.080=0.400. Location B becomes most probable even though A initially had more mass. A later observation can move the belief again. This example uses w=1 and illustrative likelihood numbers.

Quality tempering

If w=0, every likelihood becomes 1 and the posterior remains unchanged. If 0<w<1, the update is softened. If w=1, the full modeled evidence is applied. Rejected or blocked OOD measurements get zero structural evidence; caution weakens the update.

What assumptions remain

The algorithm treats sequential evidence through its approximate model; repeated correlation errors are not magically independent. The project detects duplicate payloads and penalizes redundant probing, but these measures do not prove statistical independence.

> **A clean viva answer**
> Bayesian inference does not guarantee correctness. It combines the prior and new evidence consistently within the assumed likelihood model, whose physical validity still needs testing.

Source trail: [R12] posterior_update; [R3] evidence exponent and OOD gate; [R4] duplicate rejection.

16 / POSTERIOR OUTPUTS

Reading the map and its uncertainty

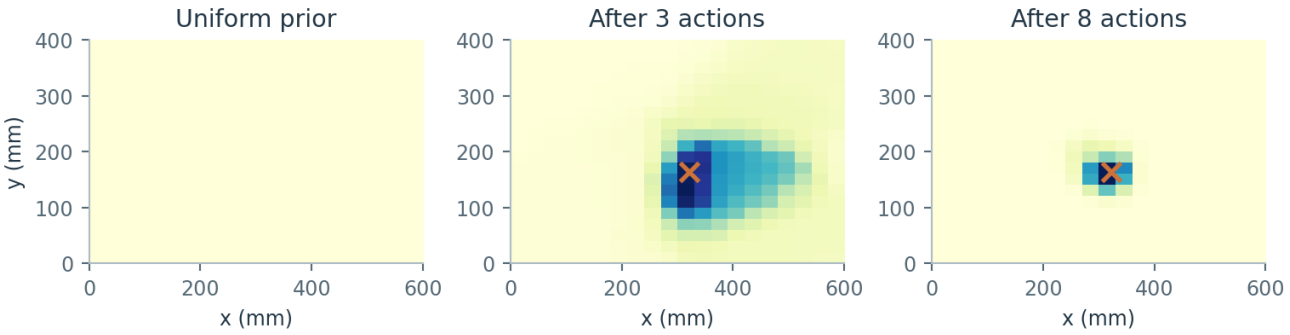


Figure 4. Current easy seed-17 run. Panels use their own color scale to show spatial shape. Orange crosses mark sealed truth only for retrospective explanation. Plot height means probability, never physical defect depth.

Output	Correct interpretation
MAP	Grid-cell center with the largest probability. This is the location used by the default revealed localization-error metric.
Posterior mean	Probability-weighted average location. With two distant modes it can lie between plausible regions.
Normalized entropy	H=-sum p log2(p), divided by log2(number of cells). Uniform belief is near 1; concentration lowers it.
90% credible area	Smallest selected high-probability set of cells accumulating approximately 90% mass; can be disconnected.
CEP50 / R90	Radius about the MAP enclosing 50% / 90% discrete posterior mass at cell centers.
Decision confidence	Composite concentration score multiplied by heuristic model trust and capped under OOD. It is not a calibrated correctness probability.

The default 20 by 20 grid has 400 cells. On a 600 by 400 mm panel, cells are 30 by 20 mm. The maximum distance from a point in a cell to its center is about 18.0 mm; this describes grid quantization, not a universal accuracy bound. A posterior can select the wrong cell by far more.

R90 can even be zero when 90% mass is concentrated at one discrete cell center. This must not be read as zero physical uncertainty inside that cell or as measured 90% field coverage.

Source trail: [R12] BeliefState.estimate; [R15] StructuralPosterior; [R9] UI containment wording.

17 / ACTIVE EXPERIMENT DESIGN

How ARGUS chooses the next experiment

A candidate experiment specifies source and receiver coordinates, frequency limits, waveform, drive amplitude, and recording duration. Candidate locations are drawn from perimeter points and representative posterior regions; the current NEO planner expands a finite waveform bank.

```
signature(z,e) = [delay, log_gain, sin(phase), cos(phase)]
phase = 2*pi*center_frequency*delay

Conceptual utility = information + hypothesis separation
                   + coverage/calibration/waveform value
                   - movement/time/energy/repetition cost
```

Core selection sequence

- Choose diagnostic, exploration, calibration, or verification mode according to structural and metrology uncertainty, trust, and history.
- Enforce amplitude, frequency, duration, spacing, model support, and no-go restrictions before execution.
- Take spatially separated representative hypotheses. The live NEO counterfactual route caps this set at five, even though the baseline configuration can request more.
- Predict a response mean and uncertainty under each remaining hypothesis. Ask how distinguishable those predicted distributions are.
- Rank feasible candidates and expose selected score components plus the top alternatives.

What the default utility really is

The NEO default uses a weighted multiobjective score with counterfactual separation, an entropy/separation information proxy, heuristic risk reduction, calibration value, trust, coverage, waveform utility, a baseline guard, costs, and redundancy. It is a ranking approximation, not exact expected posterior entropy.

> **Do not memorize a false single formula**
> The older baseline and current NEO scorer differ. Use the conceptual utility above, then say the exact configured component breakdown is visible in Reasoning and the candidate audit.

Source trail: [R5] NeoExperimentPlanner; [R16] baseline planner; [R17] counterfactual and forward contracts.

18 / PREDICT BEFORE MEASURING

Counterfactual planning: a concrete example

Suppose two remaining defect regions are both plausible. For candidate A they predict nearly identical arrival times. For candidate B they predict visibly different arrivals. With comparable physical cost, B is more useful because its observation can decide between the rivals.

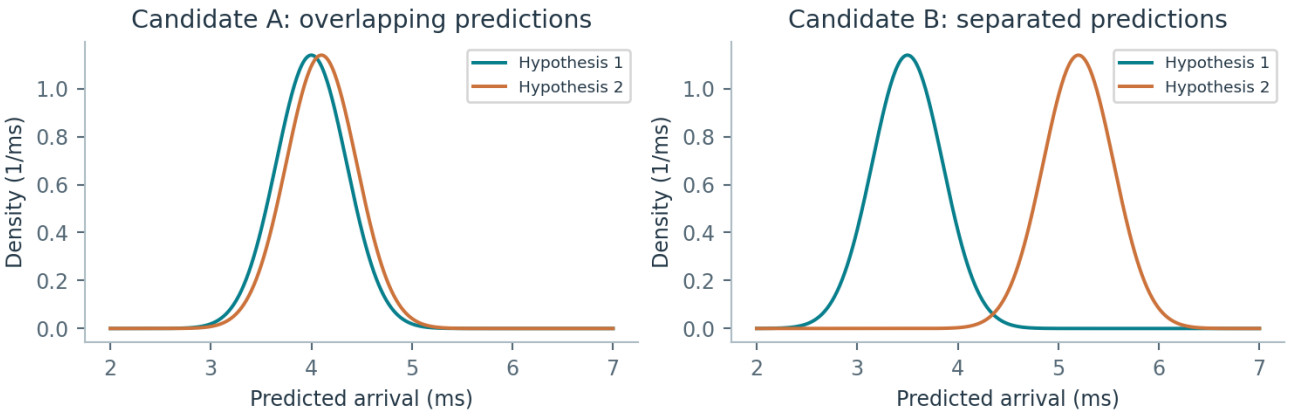


Figure 5. Teaching distributions, not an ARGUS run. Both hypotheses have 0.35 ms arrival uncertainty. The first candidate is ambiguous; the second is easier to discriminate.

Candidate	H1 mean	H2 mean	Difference / noise scale
A	4.0 ms	4.1 ms	0.1 / 0.35 = 0.29
B	3.5 ms	5.2 ms	1.7 / 0.35 = 4.86

The same difference can lose value if uncertain wave speed, timing, or placement makes the distributions much wider. NEO propagates nuisance uncertainty into prediction covariance and compares distribution overlap/separation, rather than trusting a difference of means alone.

How costs affect the decision

An informative candidate requiring a long remount or unsupported drive may be rejected or ranked below a slightly less informative feasible probe. The cost and energy numbers are configured research proxies. Amplitude squared times duration is not calibrated actuator energy in joules.

Meaning of counterfactual here
If location H1 were true, what would this controlled experiment record? If H2 were true, what would it record instead? This is forward prediction under rival hidden-state hypotheses, not an explanation of what caused the damage.

Source trail: [R17] CounterfactualExperimentEngine; [R5] utility; figure uses an explicitly illustrative Gaussian example.

19 / JOINT UNCERTAINTY

What the NEO extension adds

The original locator concentrates on a defect location grid. ARGUS NEO asks an additional question: is uncertainty coming from the object, the instrument, or model mismatch? This determines whether to diagnose, calibrate, explore, or verify.

State	Examples	Representation
Structural Z	Dominant location; approximate radius/severity/type state	Normalized 2D grid plus lightweight moments
Metrology eta	Velocity, timing, attenuation, source/receiver coupling, gain, noise, placement, temperature proxy	Bounded Gaussian parameters; velocity and timing retain a correlated 2 by 2 block
Discrepancy delta	Systematic difference between predicted and observed signatures	Online residual correction and heuristic uncertainty/trust

```
p(Z, eta, delta | data)
  approximately grid(Z | eta, delta, data)
              * q(eta | data) * q(delta | residual history)
```

This is a resource-limited approximation. Most nuisance relationships remain separate marginals; it is not full joint MCMC inference. The radius/severity/type values are not validated reconstructions, and some are initialized summaries rather than learned defect properties.

Four action modes

- **Exploration:** a broad belief needs coverage and diverse constraints.
- **Diagnostic:** rival locations need an experiment that separates them.
- **Calibration:** uncertainty or mismatch in propagation/acquisition makes further structural interpretation weak.
- **Verification:** a new diverse accepted observation challenges the leading result before a confident stop.

Only analytical and physics-signature forward levels are currently wired into the live NEO planner. Learned and offline-bank fidelity levels are extension points; do not present all four advertised levels as validated live switching.

Source trail: [R2] current disclosure; [R18] joint/nuisance state; [R19] dual control; [R17] live forward-model dictionary.

20 / METROLOGY

Calibration: do we trust the ruler?

Localization translates time into distance. If both wave speed and electronics delay are uncertain, the same observed time can arise from different combinations of them. Calibration must therefore resolve the measurement model, not just collect more defect observations.

```
Reference arrival: t = t0 + distance/v
Let b = 1/v: t = t0 + b*distance

For two distinct reference distances d1 and d2:
b = (t2-t1)/(d2-d1)
v = 1/b
t0 = t1 - b*d1
```

Example: for d1=0.18 m and d2=0.54 m, arrivals of 1.8 ms and 3.8 ms give b=0.002/0.36 s/m, hence v=180 m/s and t0=0.8 ms. Repeating only d1 improves the same constraint but cannot independently identify both unknowns.

Current implementation

The covariance state retains velocity/timing correlation. The direct-path update uses a first-order derivative H=[-L/v^2, 1] and a Joseph-form covariance update. One scalar delay counts as one observation. Varied diagnostic calibration options span distinct path lengths.

```
Arrival variance = H*C*H^T + R
Kalman-like gain K = C*H^T / (H*C*H^T + R/w)
New mean = old mean + K*(observed - predicted)
```

Reference actions update metrology and do not multiply defect evidence into the location grid. Quality alone does not measure absolute probe pose or separate source and receiver coupling. Those quantities need appropriate external measurements or controlled experiments.

> **A necessary scope distinction**
> A good direct-path reference supports propagation timing. It does not validate the entire defect-scattering model. Physical calibration requires acquired characterized reference signals; the convenience /calibrate endpoint rejects physical sessions.

Source trail: [R18] nuisance covariance; [R20] CalibrationEngine; [R4] calibrate mode rules; [R2] equations.

21 / QUALITY / OOD / VERIFICATION

Bad data, unsupported models, and stopping

Before trusting a recording, ARGUS checks sample validity and acquisition-quality proxies including signal content, clipping, repeatability, motion, and placement context. Silence, corrupt values, duplicate payloads, and rejected measurements must not accumulate unrestricted confidence.

Model state	Structural handling
NOMINAL	Use the accepted quality-tempered evidence and normal trust cap.
CAUTION	Multiply structural evidence weight by 0.35 and reduce allowable decision confidence.
OUT_OF_DISTRIBUTION	Block the structural update; request appropriate recovery evidence.
ABSTAIN	Block a confident structural result; calibration, repair, or escalation may be needed.

OOD means the observation does not fit the current empirical model envelope. The code combines robust residual distance, conformal-style nonconformity when reference data suffice, ensemble disagreement, a physical acoustic reference score, and quality penalty. These are fallible guards.

Order matters

Diagnostic OOD is assessed against the belief **before** fusion. Otherwise an outlier could first move the posterior toward itself and then appear consistent. A caution state weakens evidence; OOD or abstention gives zero structural evidence.

Stop reasons

- **STOP_CONFIDENT:** concentration/risk conditions are supported by the latest accepted, distinct, nominal, spatially consistent verification observation.
- **STOP_BUDGET:** measurement budget exhausted; this does not mean the location is reliable.
- **STOP_LOW_VALUE_OF_INFORMATION:** another modeled observation has low expected utility.
- **STOP_OOD:** the current model cannot support confident interpretation.

> **Do not equate a guard with proof**
> OOD can miss mismatch and can reject nominal data. Verification is a consistency test, not a physical safety certificate or calibrated field coverage claim.

Source trail: [R3] process_signal/status; [R6] stopping; [R21] OOD and quality code; [R2] current audit.

22 / ADVANCED METHODS

The two experimental planning options

1. Nuisance-aware arrival-time information

This option coarsens the complete spatial posterior into regions and represents predicted scalar arrivals as a Gaussian mixture. Propagation variance includes the retained velocity/timing cross term as well as placement and other assumed uncertainty.

```
p(y | e) = sum_h p(h) * Normal(y; mu_h(e), var_h(e))
I(H;Y | e) = E[log2(p(y|h,e)/p(y|e))]
```

Gauss-Hermite quadrature estimates the information about regional identity in bits. It is more explicit than an overlap proxy, but only for its approximate arrival model. It is not exact information about the full waveform or every cell.

2. Calibration-aware risk planning

The optional risk policy compares diagnostic and calibration actions in common squared-position-error units. Squared-error loss is minimized by the posterior mean. The policy retains within-region spread so grouping cells does not erase localization uncertainty.

```
R(p) = E[||x - posterior_mean||^2]  [m^2]
Value(e) = prior risk - expected terminal risk - cost
Default cost conversion = 0.0001 m^2 / research cost unit
```

It branches over possible first observations, conditions the second prediction using shared nuisance covariance, chooses a second action within each branch, and can stop after the first. Actual lookahead is at most two measurements with a finite candidate beam.

What to say about horizon and performance

The fast legacy horizon reranker is a geometric route-score proxy with no observation branching. The newer risk option has observation branches. Both experimental flags are off by default, and their saved evaluations do not establish general accuracy superiority.

> **Why the distinction matters**
> More explicit mathematics is useful research, but model mismatch and computation cost can still make a policy perform worse. Page 28 includes the negative and inconclusive results.

Source trail: [R22] nuisance_design.py and bayes_design.py; [R2] current disclosure; [R23] research evaluation artifacts.

23 / OPTIONAL FORWARD SURROGATE

Where machine learning fits

The neural model learns **state + experiment -> response features**. It is a forward surrogate, not a direct classifier that maps a waveform to the final defect coordinates. The main live inference path remains physics-based.

```
24 inputs -> Linear(96) -> SiLU -> LayerNorm
          -> Linear(96) -> SiLU -> Linear(15 outputs)
```

Input group	Count / examples
Defect state	5 values: x, y, radius_x, radius_y, severity
Type encoding	4 one-hot backend defect types
Probe geometry	4 source/receiver coordinates
Excitation	2 normalized frequencies, amplitude, and 3 one-hot waveforms
Panel / material	Width, height, normalized velocity, attenuation, and resonance

The 15 targets include amplitude statistics, timing, spectral statistics, band energies, decay, and SNR. Although the analysis displays a separate noise estimate, it is not one of these 15 training targets. Noise, damping, and instrument delay are randomized but not explicit in all inputs, leaving irreducible predictive variation.

Training procedure

Generate domain-randomized examples; split data 70/15/15; fit standardization on training data; train with Smooth L1 loss and AdamW; monitor validation loss with early stopping; then evaluate the untouched test split. Group correlated measured channels by physical scenario or specimen to prevent leakage.

Saved artifact	Held-out evidence
15,000-example reference	Mean feature R-squared 0.47567; standardized MAE 0.45690.
50,000-example candidate	Internal mean feature R-squared 0.49288; saved separately with additional comparison artifacts.

A precise ML statement
These metrics quantify response-feature prediction on held-out synthetic examples. They are not 47.6% or 49.3% defect-location accuracy. The live NEO planner currently exposes forward levels 0 and 1, not a validated learned-surrogate adapter.

Source trail: [R24] generator/trainer and ForwardSurrogate; [R25] checkpoint metadata and separate training report.

24 / MEASURED-DATA EVIDENCE

Data sources and the simulation-to-reality gap

Source	What it contributes	What it does not prove
KU Leuven LMSD	Measured CFRP multistatic frequency-response data and known added-mass scenarios.	Accuracy on arbitrary defects or the eventual phone apparatus.
TU Darmstadt GFRP	2,568 recorded taps from 8 physical plates; pointwise intact/defect labels.	Adaptive 2D bistatic localization.
Bologna / Zenodo	Simulated propagation paths plus 9 measured impact positions and 3 channels on one aluminum plate.	Persistent hidden-defect localization across specimens.

The GFRP experiment

The repository converts waveforms to the ARGUS feature contract and evaluates a gain-robust detector with complete physical plates held out. Saved pooled results: balanced accuracy **0.775**, ROC-AUC **0.911**, and F1 **0.634**. Grouping matters: random splits of repeated taps can leak specimen identity into test data.

The impact experiment

The saved simulation-transferred localizer has mean error **0.175 m**. Leave-one-position-out calibration on the other eight measured locations reduces it to **0.141 m**. This is one plate and transient impact localization. An affine timing correction increased error to 0.197 m and was rejected.

The Zenodo publisher describes a 1 by 1 by 0.003 m plate, three sensors, 2 MHz acquisition, and 250 kHz excitation [W4]. That measurement domain is very different from the main 16 kHz/audio demonstration.

Runtime bridge

A measured GFRP acoustic reference adds an OOD penalty for physical input outside the reference envelope. Feature-transport fitting aligns simulated and measured statistics. Moment alignment is not by itself proof of generalization to an unseen device.

> **Your defensible evidence statement**
> We have a functioning simulated adaptive loop and separate measured-data bridging experiments. We still need a blind physical campaign of the complete adaptive loop on our actual apparatus.

Source trail: [R26] SIM2REAL_REPORT and metrics; [R27] real-data adapters; [W4] publisher dataset.

25 / ACQUISITION

Physical hardware, phone, and two-laptop operation

Acquisition path	How it works
Simulation	A seeded local twin returns a waveform from the sealed defect scenario.
WAV upload	PCM/float samples are decoded in memory, integer channels normalized, stereo averaged, and resampled to the configured rate.
Browser microphone	The browser requests microphone access and produces a WAV/sample recording for the selected experiment.
Local microphone	sounddevice acquires from the operating-system audio input.
ESP32 probe	An external driver excites the object; the conditioned analog signal is acquired and sent through a serial protocol.
Phone / Laptop B	/probe or run_edge_node.py transmits acquisition and metadata to the main local backend over the LAN.

Reference ESP32 wiring

GPIO 25: driver control for the exciter.
GPIO 34: protected, conditioned analog input in the 0-3.3 V range.
GPIO 2: status LED.
GND: common ground for compatible sensor/driver circuits.

A piezo front end requires buffering, protection, clamping, and biasing; the firmware guide uses a midpoint around 1.65 V. The actuator needs a suitable external supply and transistor/MOSFET driver. This is a reference acquisition design, not evidence that hardware was built or tested.

Data and synchronization limits

The serial connection is 115200 baud; buffered acquisition can occur at 16 kHz even though text transfer is slower than real time. The phone supplies position/orientation/motion and timestamps where browser permissions allow. These metadata do not establish synchronized clocks or measured coupling force.

Physical inference needs known excitation timing, repeatable geometry, healthy/characterized references, controlled mounting, and a sensor response appropriate to the structure. A casual microphone recording does not automatically meet that contract.

Source trail: [R28] hardware/firmware guide; [R29] phone/distributed docs; [R4] upload/resampling validation.

26 / SUPPORTING SYSTEMS

Interface, persistence, and evidence records

What each interface view contributes

Overview: posterior, source/receiver recommendation, uncertainty, and candidate ranking.
Reasoning: hypotheses, modeled outcomes, metrology, trust, utility, and alternatives.
Signals / Evolution: processed measurements and belief history.
Benchmark / Research: saved and queued studies.
Assurance / Evidence: sensor status, stop controls, ledger checks, and bundle export.
Registration: camera guidance for executing normalized coordinates.

Camera mapping

```
[image_u, image_v, 1]^T ~ H * [panel_x, panel_y, 1]^T
```

The main Registration component solves an eight-parameter planar homography from four clicked panel corners. It overlays guidance and the map on camera coordinates. The phone probe separately uses a simpler four-corner bilinear mapping. Neither is an automatic calibrated 3D pose reconstruction.

Persistence and reproducibility

SQLite stores session state, measurement history, compressed raw float32 signals, events, models, and local research jobs. The browser remembers only the last session identifier for convenience. The simulation truth is withheld from public state until reveal; physical reveal is unavailable.

Evidence ledger

```
record_hash = SHA256(canonical record content)
Each record includes previous_record_hash
+ waveform/state/configuration/model/seed information
```

Verification checks recorded content and hash-chain linkage. An independently retained head hash and count can detect a recomputed alternate history. Without an independent checkpoint or signature, a rewritten full chain can still be internally consistent. This is not a blockchain, signature, or proof that the sensor truth is correct.

A single-worker SQLite research queue supports benchmarks, calibration/ablation studies, banks, demos, and training. Constraints and a latched emergency stop govern the software session; the LAN remains a trusted local deployment.

Source trail: [R9] UI; [R30] camera mapping; [R31] repository, jobs, ledger, and bundles.

27 / RESULTS / VERSION BOUNDARY

The historical benchmark: what it supports

The saved baseline study used 30 identical seeded medium cases, at most 10 experiments, and random, uniform-grid, and active ARGUS policies. These numbers belong to that saved controller version, not every later NEO change.

Policy	Mean error	Final entropy	Success <=15 mm
Random	13.38 mm	0.498	56.7%
Uniform grid	13.26 mm	0.621	60.0%
ARGUS active	12.33 mm	0.419	73.3%

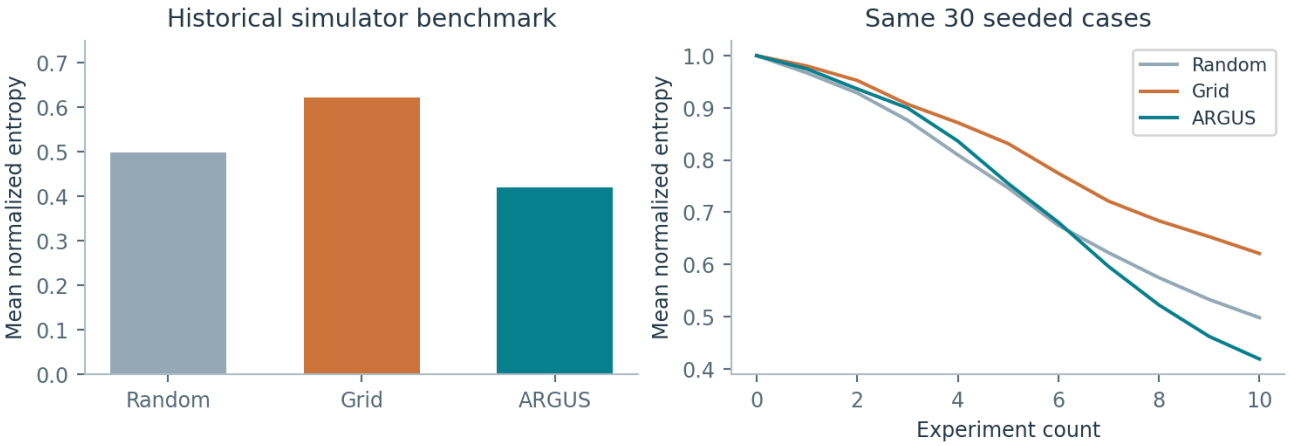


Figure 6. Saved historical simulator benchmark, 30 paired cases. Lower modeled entropy measures concentration of the posterior, not physical correctness by itself.

Paired entropy advantage over random: **0.079**, bootstrap 95% interval **[0.063, 0.095]**. Over grid: **0.202**, interval **[0.180, 0.224]**. These intervals exclude zero within the declared simulator study.

Mean error advantage over random: 1.05 mm, interval [-0.67, 2.81] mm. Over grid: 0.93 mm, interval [-1.02, 2.76] mm. Both include zero, so the saved study does not establish a statistically clear error improvement.

A review-safe result sentence

In the saved historical simulator benchmark, the adaptive policy produced lower posterior entropy and higher observed 15 mm success rates. Error superiority and physical field accuracy were not established.

Source trail: [R32] benchmark_results/benchmark.json. Values read directly from its summary and paired comparison fields.

28 / RESULTS / CURRENT CODE

Current experimental evidence and fresh checks

Calibration-aware risk policy evaluation

The saved audit study compares legacy and risk policies on six paired seeds per condition, six measurements, and shared diagnostic opportunities. It reports 24 completed synthetic runs; source_changed_during_run=false. This is a small engineering evaluation.

Condition	Legacy mean error	Risk mean error	Risk advantage / 95% CI
Nominal	43.90 mm	12.58 mm	31.32 mm [-2.71, 93.38]
Mismatch + noise	150.22 mm	203.78 mm	-53.56 mm [-174.99, 17.82]

Both intervals include zero. Both policies succeed within 20 mm in 5/6 nominal cases and 0/6 stressed cases. Risk computation averages about 21 seconds per six-measurement case versus about 0.35-0.39 seconds for legacy. No overall superiority is established.

The separate nuisance-aware study has 80 synthetic runs: nominal errors 15.12 vs 15.10 mm, stress errors 180.14 vs 146.56 mm. Paired intervals include zero; both methods fail the 30 mm threshold in 17/20 stress cases. Both options remain experimental.

Checks run for this guide, 7 October 2026

Fresh check	Observed outcome
scripts/doctor.py	Readiness true; required imports, SQLite initialization, and one closed-loop measurement pass.
Backend suite	110 collected tests; full suite completed successfully. One upstream Starlette/httpx deprecation warning.
Frontend tests / types	24 tests passed; TypeScript no-emit check passed.
Current easy seed-17 demo	8 actions; about 10.0 mm MAP error; final normalized entropy 0.473; STOP_BUDGET.

Evidence hierarchy
Software checks establish behavior of the code. A one-seed demo illustrates it. Paired simulator studies compare policies within a model. None substitutes for a blind physical localization study.

Source trail: [R23] audit and nuisance studies; fresh commands executed in this workspace. Current demo trace retained with the guide QA files.

29 / REVIEW WALKTHROUGH

Run the demo and explain what you show

Start from the project root

```
python scripts/doctor.py
.\run_argus.ps1

Browser: http://localhost:5173
ARGUS API in this workspace: http://localhost:8001
API documentation: http://localhost:8001/docs
```

frontend/.env.local selects port 8001 here; the general code default is 8000. The launcher propagates the configured port, and the browser checks ARGUS service identity. Use the local launcher output if your environment changes.

A three-minute demonstration

- **0:00-0:25:** state the problem. Start a simulation with experimental toggles off. Point out that ground truth is locked.
- **0:25-1:20:** run several actions. Show changing probe geometry, waveform, probability field, and the reason for the next probe. Identify any calibration action explicitly.
- **1:20-1:50:** open Signals and explain measured response, spectrum, and baseline-subtracted matched filtering. Open Reasoning once to show alternatives.
- **1:50-2:15:** reveal the simulation truth only after the measurement sequence; quote error in mm and the actual stop reason.
- **2:15-2:40:** show Evolution or the saved Benchmark, stating its evidence source and version boundary.
- **2:40-3:00:** describe the physical acquisition path and the next blind validation experiment.

Browser-independent fallback

```
python scripts/demo_simulation.py --preset easy --seed 17 --experiments 8
```

This exact command was checked for the guide: eight actions and about 10 mm error. The UI uses a random seed unless supplied through its API, so a new UI session is not guaranteed to repeat that trajectory.

> **If the map looks uncertain**
> Read the action type and stop explanation. Calibration or abstention is part of the controller behavior. A budget stop must be presented as a budget stop, even when the revealed position looks good.

Source trail: [R33] launcher, env setting, demo script, and docs/DEMO_GUIDE; [R9] UI session creation.

30 / SPEAKING NOTES

Your Review 1 presentation script

Thirty-second opening

Memorize this

Our project is ARGUS, Adaptive Recursive Guided Uncertainty Sensing. It locates one dominant hidden defect using vibration responses and a Bayesian probability map. Before each measurement, it predicts how competing defect locations would respond to possible probe placements and waveforms, then chooses an informative feasible experiment. The recorded response updates the same belief, and the loop continues with calibration, verification, or stopping when appropriate.

A five-minute structure

Time	What to say / show
0:00-0:45	Problem: opaque structures, uncertain defects, and limited measurements. Objective: make acquisition adaptive.
0:45-1:30	Physics: source-defect-receiver path; equal-delay ellipse; several geometries narrow ambiguity.
1:30-2:15	Architecture: React instrument, FastAPI engine, simulator/sensor inputs, Bayesian fusion, planner, and SQLite evidence.
2:15-3:15	Method: baseline subtraction, matched filtering, quality-weighted Bayes update, and counterfactual candidate ranking.
3:15-4:15	Demo: one evolving map, changed recommendation, signal evidence, and sealed-truth reveal.
4:15-5:00	Evidence and next work: historical simulator uncertainty reduction; separate measured-data bridge; blind physical campaign still required.

Suggested slide order if you need one

1. Title and problem. 2. Aim and scope. 3. Source-defect-receiver diagram. 4. Closed-loop block diagram. 5. Signal processing and Bayes update. 6. Adaptive planner and NEO calibration. 7. Demo/results with source labels. 8. Limits and next milestone.

A strong closing sentence

The implemented contribution is a working, auditable loop joining physical response prediction, sequential measurement choice, and recursive inference. Our next milestone is to test whether this loop improves localization under equal budgets on independently held-out physical panels.

Source trail: Synthesized review script grounded in [R1]-[R6], [R23], and [R32].

31 / QUESTIONS 1-12

Viva practice: the essential answers

1. What does ARGUS stand for?

Adaptive Recursive Guided Uncertainty Sensing. Adaptive means the next probe changes with evidence; recursive means new evidence updates the existing belief.

2. What exactly is the problem?

We do not know the interior defect location and have limited measurements. We choose where and how to excite and record in order to reduce the remaining ambiguity.

3. What is the main output?

A probable location, uncertainty summary, recommended next experiment, rationale, and measurement history. Simulation can additionally reveal truth for scoring.

4. Why does one measurement not locate a point?

One bistatic arrival constrains a sum of distances. Many points share that sum and form an ellipse; different geometries add new constraints.

5. Why is this called AI if physics is used?

The decision system maintains uncertain hypotheses and chooses actions through probabilistic inference and adaptive planning. A neural network is an optional forward-model component.

6. Is this reinforcement learning?

The live controller uses model-based Bayesian-style experimental design and configured utility. It is not a trained reinforcement-learning policy.

7. What is a digital twin here?

A computational model predicts responses to proposed probes. Our live twin is a reduced physical model, not validated finite-element reconstruction.

8. What is the Bayesian update?

Multiply the existing spatial belief by the new quality-tempered likelihood and normalize. The result is the prior for the next measurement.

9. What is matched filtering?

Correlate the defect residual with the known excitation. A delayed matching response produces correlation support at its propagation lag.

10. What is entropy?

A measure of how spread the probability distribution is. Lower entropy means more concentration under the model, not automatically greater correctness.

11. What is the planner selecting?

A feasible combination of source, receiver, frequency band, waveform, amplitude, and duration, plus an action mode such as calibration or verification.

12. What is your tech stack?

Python with NumPy/SciPy/FastAPI/SQLite; React/TypeScript with Vinext/Vite; optional PyTorch and scikit-learn; serial/audio and ESP32 acquisition support.

32 / QUESTIONS 13-24

Viva practice: the difficult questions

13. How is the next experiment chosen?

Predict outcomes under rival locations, compare their separation and uncertainty, balance configured costs, enforce constraints, and select the highest-ranked candidate in a finite bank.

14. Is expected information gain exact?

The default is an entropy/separation proxy. An optional method integrates information under a coarsened Gaussian arrival model, which is still an approximation to the physical observation.

15. Why calibrate instead of diagnose?

Uncertain speed or timing can make defect explanations indistinguishable. A reference may improve subsequent diagnostic prediction. It must not be counted as defect evidence.

16. Can one reference estimate speed and delay?

Not independently. One time gives one constraint on two unknowns; different path lengths or other measured information are required.

17. What is the ML model doing?

A 24-input MLP predicts 15 response features. It is optional and is not the default waveform-to-location inverse model or a validated live NEO fidelity adapter.

18. What accuracy can you claim?

State a specific study: the historical 30-case simulator gives 12.33 mm mean active-policy error. Its paired error intervals cross zero, and physical adaptive accuracy is not established.

19. What if a microphone records room noise?

Quality and model-support checks can reject or weaken the update, but they are fallible. Controlled excitation timing, references, coupling, and measured validation are still required.

20. Can it handle multiple defects or depth?

The current inverse core focuses on one dominant localized scatterer. Candidate modes and summary moments are not a validated joint multi-defect or depth solution.

21. How do you know the planner did not see truth?

Public state hides truth until reveal, planning consumes belief/model state, and the acquisition simulator is separate. Software tests check truth sealing; hidden truth is used for scoring.

22. Is the ledger tamper-proof?

It checks content and hash linkage. An independent checkpoint improves detection. An attacker replacing the entire unanchored history can recompute hashes, so it is not tamper-proof.

23. Is this a new invention?

The building blocks have prior art. We can describe our implemented integration and experimental policies, but novelty or patentability has not been established.

24. What is your next concrete experiment?

Acquire controlled healthy/defective recordings on the actual apparatus, retain all candidate outcomes, hold out whole specimens/locations, and compare adaptive and fixed policies at equal budgets.

33 / NEXT MILESTONE

Limits, failure modes, and a credible roadmap

Failure / limit	Why it matters	Next evidence or change
Incorrect speed / timing	The wrong ellipse can fit the observation.	Varied-distance references and held-out calibration checks.
Coupling or mounting drift	Amplitude and timing change without a new defect.	Repeatable contact fixtures; before/after healthy references.
Boundary / dispersive modes	Unmodeled arrivals mimic scattered evidence.	Characterize dispersion and reflections; improve the model.
Multiple defects	One-scatterer likelihood can fuse incompatible paths.	Validate one defect first, then a joint multi-defect model.
Coarse grid	Reported MAP and R90 are discretized.	Resolution studies and adaptive/subcell inference with validation.
Concentrated wrong belief	Low entropy can hide model error.	Predictive checks, diverse verification, and coverage studies.
Public-data domain mismatch	A new device/material differs from the training setup.	Blind measured tests on the actual apparatus and unseen specimens.

A review-ready validation plan

- Prepare a characterized healthy specimen and known reversible anomalies; measure geometry independently and use repeatable mounting.
- Record every feasible candidate outcome needed for fair offline policy replay, including explicitly labelled reference measurements.
- Separate entire specimens or anomaly locations into development and blind test groups; keep repeats within their group.
- Compare adaptive, random, and fixed policies using equal measurement opportunities and declared costs. Retain failed and abstained cases.
- Report location error versus budget, success at declared tolerances, R90 empirical coverage, rejection/abstention rates, and computation cost.

Tonight's priority
Be able to draw the loop, derive the ellipse, explain one Bayes update, locate the key code, state one result with its scope, and describe this next experiment. Those six abilities matter more than memorizing every implementation detail.

Source trail: [R2] current support matrix; [R26] measured-data limits; [R34] physical_campaign.py and validation plan.

34 / QUICK REFERENCE

Glossary and notation

Term	Meaning in this project
NDT / NDE	Nondestructive testing / evaluation: inspect without intentionally destroying the specimen.
SHM	Structural health monitoring; an intended broader application area.
Bistatic path	Source-to-defect plus defect-to-receiver path with potentially different source and receiver.
Prior / likelihood / posterior	Previous belief / compatibility of new data with a hypothesis / updated belief.
MAP / posterior mean	Most probable grid cell / probability-weighted average location.
Entropy	Probability-distribution spread, reported in bits or normalized units.
Credible region / R90	Modeled posterior mass region / MAP-centered radius enclosing 90% discrete mass.
Counterfactual	Predicted observation if a particular hypothesis and controlled experiment were used.
Nuisance / metrology	Instrument and propagation variables that affect the observation but are not the target defect location.
Model discrepancy / OOD	Prediction mismatch / evidence outside the current empirical model envelope.
Surrogate / domain randomization	Learned approximation of the forward response / variation of simulated conditions during data generation.
Ablation / leakage	Remove a component to test its effect / related information contaminating train-test separation.
Paired bootstrap interval	Resample matched cases to estimate uncertainty of a policy difference within the study.
Hash chain / checkpoint	Records linked by content hashes / an independently retained anchor for verification.

```
S, R, z : source, receiver, candidate defect position
L : path length (m)        v : velocity (m/s)
t0 : instrument delay (s) fs : sampling rate (Hz)
u(t), y(t) : excitation and observed response
w : structural evidence weight
D : plate bending rigidity; h : thickness
E, rho, nu : modulus, density, Poisson ratio
```

Source trail: Definitions aligned to the current implementation and this guide; these are not certification terms.

35 / REFERENCES

Sources and reproducibility trail

Repository evidence inspected on 7 October 2026 in C:/Users/ripan/Hackathon_proj. Existing uncommitted changes were included; this guide describes the working tree rather than a frozen release. Current code and the current support disclosure take precedence over historical architecture files.

Key	Local source (relative to project root)
R1	README.md; docs/PROJECT_KNOWLEDGE.md; docs/SCIENTIFIC_MODEL.md
R2	docs/CURRENT_INVENTION_DISCLOSURE.md; docs/AUDIT_REMEDIATION.md
R3-R6	backend/app/services/engine.py; session_manager.py; active_learning/neo_planner.py; decision/stopping.py
R7-R9	backend/requirements.txt; frontend/package.json; backend/app/main.py; frontend/app/page.tsx; hooks/useArgusSession.ts; services/api.ts
R10-R12	backend/app/simulation/physics.py; frontend/utils/virtualAcoustics.ts; backend/app/inference/belief.py
R13-R17	backend/app/core/config.py; safety/constraints.py; signal/processing.py; inference/structural_posterior.py; active_learning/planner.py; counterfactual.py; digital_twin/forward_models.py
R18-R22	backend/app/inference/joint_state.py; nuisance_posterior.py; calibration.py; diagnostics.py; control/dual_control.py; ood/detection.py; active_learning/nuisance_design.py; bayes_design.py
R23	research_results/audit_risk_validation.json; nuisance_design_validation.json; docs/RESEARCH_EVALUATION.md
R24-R25	scripts/generate_dataset.py; train_model.py; backend/app/models/surrogate.py; models/forward_surrogate*.json; docs/SURROGATE_TRAINING_2026_09_20.md
R26-R29	docs/SIM2REAL_REPORT.md; models/sim2real_metrics.json; ae_sim2real_metrics.json; scripts/prepare_*_dataset.py; docs/HARDWARE.md; PHONE_PROBE.md; DISTRIBUTED_DEMO.md; firmware/esp32/README.md
R30-R31	frontend/components/CameraOverlay.tsx; frontend/app/probe/page.tsx; backend/app/database/repository.py; evidence/ledger.py; bundles.py; research/jobs.py
R32-R34	benchmark_results/benchmark.json; run_argus.py; run_argus.ps1; scripts/doctor.py; demo_simulation.py; physical_campaign.py; docs/DEMO_GUIDE.md

Primary external references

[W1] SciPy: correlate and matched filtering

[W2] COMSOL: Lamb waves for plate inspection

[W3] COMSOL: eigenfrequency and plate stiffness context

[W4] Zonzini, Donati, De Marchi: numerical/experimental impact dataset

External references support the cited background or publisher dataset description; project metrics are from local artifacts. Fresh checks: doctor; full backend tests (110); frontend tests (24); TypeScript; seed-17/easy/8-action demo. A production build and interactive browser QA were not rerun for this guide.